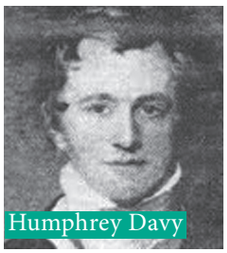
INTRODUCTION

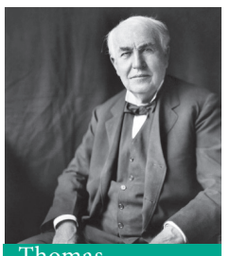
We are experiencing many hoardings, advertising displays, high definition television in our daily routine life. Of course, these devices have very good features with long life and attractive characteristics. Do you have any idea about these devices and their origin? Most of these devices are built around a special type of semiconductor devices called LED, LDR, LCD, photodiode and photo transistor. They are opto-electronic devices.

With these devices we study about the working of FET, MOSFET, SCR, DIAC, TRIAC, IGBT and their application.

History of Electric bulb


Humphrey Davy

The first electric bulb was made in 1800 by Humphrey Davy. When he connected wires to the battery and a piece of carbon, the carbon gets glowed and producing light. It was known as electric arc lamp, but failed to glow for a long time.


Thomas Alva Edison

The inventor Thomas Alva Edison (USA) discovered a carbon filament in an oxygen free bulb glowed for over 1500 hours. Before this, he met hundreds of failures in his experiments. He discovered incandescent light bulb on 1879.

Edison and Tesla experimented with florescent lamps in 1890s, but Peter Cooper Hewitt successfully made fluorescent lamp in the early 1900s.


Nick Holonyak

Edward E. Hammer from General Electric Company invented the spiral compact fluorescent lamp invented in the year 1976. The first practical LED was created by Nick Holonyak in 1962. It emitted light in the visible part of the frequency range. It was red LED. In 1972, M-George Craford invented the first yellow LED and a brighter red LED.

6.1 LIGHT EMITTING DIODE (LED)

T4H3K4

LED is a PN junction device, which emits light when forward biased, by a phenomenon called electroluminescence. In all semiconductor PN junctions, some energy will be radiated as heat and some in the form of photons.

LEDs are widely used in indicators, smart phones, digital watches, solid state video displays, calculators, digital computers, electronic panels and optical communication system.

Construction

Figure 6.1 shows the construction of LED and its symbol. Here, an *N* type layer is grown on a substrate over which a *P* type layer is deposited by the process called diffusion. The metal anode connection is made at the outer edges of the 'P' layer. It provides more central surface area for the light to escape.

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